

THE GEMINI PROTOCOLS — CHAPTER 01

PHASE 01: STARSHIP ENGINE: THE ENGINE CHAIN —: THE 200 M × 20 M QUINN TURBINE, THE LONG JOHNS, THE CHEMICAL SPIN-UP PAIRS, THE HYDROGEN FACTORY, THE DISTRIBUTED CRYO KEEL (SEVENTEEN DROP-DOWN TANKS), THE BABUSHKA STORE, THE MICROWAVE-HYDROGEN MAIN ENGINE, THE HELIUM PLASMA CORES, THE PLASMA ENGINES, THE TETHER CAPTURE ARRAY, AND THE SHEARER FLEET (40 MW CRUISE / 150 MW EMERGENCY, CARBON SHIP-SIDE, STEEL COLONY-SIDE)

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat 3 AI (The Initial Interstellar Co-Engineering Alliance). License: CERN Open Hardware Licence v2 (CERN-OHL-W-2.0). Project registry: hackaday.io/project/206187. Chapter v5.21 — 4 September 2026
STATUS: CURRENT

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

This chapter is written for idiot-proofing — for AIs and for speed. Rules: (1) Build exactly what is written here; every figure is seated law from the master record. (2) Where this phase references another phase, THE DIRECTIONS FROM THAT PHASE ARE INLINED IN §7 — you never leave this document mid-build. (3) Dead ends are listed in §8 as DO NOT BUILD so you do not waste time re-deriving them. (4) Check your work against §9 — if your numbers disagree with §9, your work is wrong, not the chapter. (5) Change control is §10: only the Author's orders change this design. (6) this chapter carries current doctrine only — the audit trail lives in the master record, never in these pages.

§1 — WHAT YOU ARE BUILDING — THE CHAIN

Nine machines and one fleet, in the order the ship uses them. The QUINN TURBINE is the battery — it stores the power when required. The LONG JOHNS hold the turbine — the magnetic point-bearing set; nothing touches — and they lower each engine section through its bay doors for spin-up and retract it after. The SPIN-UP PAIRS spin the turbine — chemical rockets on the rim, the only engines that can live there. The HYDROGEN FACTORY turns power and water into hydrogen. The CRYO HULLS carry the ship's hydrogen — seventeen drop-down tanks along the keel. The BABUSHKA STORE holds Excess to the Ships

stores from Gas Giant Harvesting. The MAIN ENGINE is the microwave-hydrogen machine (plasma)— it throws the Earth departure and the Jupiter run-up, and drives the deep-space legs. The Helium Machines Plasma Cores are— shields and tapping, The PLASMA ENGINES are the terminal machine. The TETHER CAPTURE ARRAY is the energy harvester at the giants. The SHEARER FLEET helps collect at the stops pushing to the collector Array and funnel. The FORWARD FRAME SECTION is the chassis-linked steel frame at the forward lower deck between the cryo rooms — +1,000 t of steel.

THE INTENTIONS, STATED ONCE AND BINDING ON EVERY SECTION: HYDROGEN DOES THE WORK — the main engine burns pure hydrogen under microwave fire, no oxygen aboard the thrust path. HELIUM DOES THE WATCHING — the plasma cores carry the shields and the tapping. THE ROTOR IS THE BATTERY — the Quinn turbine is a flywheel, not a source; the payer in the dark is the store. THE TETHERS ARE THE ENERGY HARVESTERS — at the giants The SHEARER FLEET helps collect Hydrogen and helium where applicable, the ship drops tether wires into nuclear-level plasma and takes the giant's rotation as energy, and the shearers take its sky as propellant. THE LONG JOHNS HOLD THE ROTOR — both-north magnetic points, position and coil-gap datum in one set. CHEMICAL ONLY ON THE RIM — the spin-up pairs carry their own power and are indifferent to their own exhaust. STATION KIT NEVER SHIP KIT — the heavy harvest infrastructure stays at the giant.

THE CHAIN AT A GLANCE

+--> [6] PLASMA CORES (Phase 11) — HELIUM machines.

| shields + tapping; cruise regime; collection at the gas giants

|

[8] TETHER ARRAY — the harvest at the giants (Jupiter first): drop the wires, take the rotation

| power

v

[1] QUINN TURBINE (200 m long x 20 m diameter, $r = 10$ m — the battery:

960 MWh high line at 625 m/s rim; 150 MW anchor at 236 m/s; 40 MW cruise)

| held by [7] THE LONG JOHNS — both-north magnetic point bearings, nothing touches

| spun by [9] THE SPIN-UP PAIRS — RS-25-class chemical rockets on the rim, CHEMICAL ONLY

| power

+--> [2] HYDROGEN FACTORY (electrolysers, 50 kWh/kg — 19.2 t/day @ 40 MW, 72 t/day @ 150 MW)

| hydrogen

v

[3] CRYO HULLS (17 drop-down tanks along the keel, 25 m drop, ~340 m keel span — 19,362 t; the payers in the dark)

|

+--> [4] MAIN ENGINE — THE MICROWAVE-HYDROGEN MACHINE (throws the climb,
| the Earth departure and the Jupiter run-up; pure H₂, no oxygen)

|

+--> [5] PLASMA ENGINES (Phase 11 supplementary act) — the terminal machine;
electromagnetic thruster arrays; magnetic nozzles, no wall contact

[10] SHEARER FLEET — the stop-time collection shears to the ships Array and funnel: at the gas giants, collect HYDROGEN FIRST,

refill the stores, then drop the tethers (the operational variant, §10).

§2 — MACHINE ONE: THE QUINN TURBINE

§2.1 Dimensions and ratings (seated figures)

- Rotor: 200 m long × 20 m diameter ($r = 10$ m), four 50 m sections — the '200 m x 20 m Quinn turbine'.

- Rating: 40 MW cruise / 150 MW emergency. The emergency figure was dropped from 200 MW to 150 MW by the Author's order 'for everyone to easily choose steel or carbon no redlines visible.' 150 MW is FINAL — do not design to 200.

- Spin anchor: full spin $625 \text{ m/s} = 596.8 \text{ rpm} = 62.5 \text{ rad/s}$ on the 20 m-diameter rim (rotation period 0.1005 s; rim acceleration 3,983 g). Stored energy 960 MWh — service windows: full discharge 24 h at the 40 MW draw, 6.4 h at 150 MW; to the 50% speed floor (720 MWh usable), 18 h at 40 MW and 4.8 h at 150 MW. Lenz reaction torque 0.64 MN.m at 40 MW, 2.40 MN.m at 150 MW — drawing power slows the rotor, and the slowdown IS the fuel gauge (Phase 249 doctrine).

§2.2 The rim ruling — materials law

- SHIP-SIDE: extruded carbon-composite rim. At full spin (625 m/s): carbon hoop stress 625 MPa against a 1,250 MPa working ceiling — margin 2.0×, no redline.

- COLONY-SIDE: steel is the spec, run at the steel anchor — $236 \text{ m/s} = 225 \text{ rpm}$ on the 20 m-diameter rim (the 150 MW anchor): steel hoop stress 436 MPa against a 1,000 MPa working ceiling — margin 2.3×, no redline, storing ~137 MWh per rim. Steel does not take the ship's 625 m/s spin; the colony machine

spins slower and stores less per rim — a colony machine is a replaceable item; replace where you can, over-margin where you can't.

- WHY (the Author's life-cycle logic): a colony machine is a replaceable item; the ship's machine is too big for ship replacement — replace where you can, over-margin where you can't. Either material, either yard, no redline in sight.

§2.3 The magnet tubes — rotor construction

- Extrude the outer ring WITH tube channels; slide the magnets straight in, bare-faced — seated as the only mechanically sound way to face them.

- CONSTRAINT: the field size chosen MUST reflect the carbon's impedance — carbon fibre conducts; a careless field will eddy-heat the rim. Size the field to the rim material.

- LAYOUT LAW: the ROTOR carries the magnets; the STATORS stay small and are seated in the outer hull wall — dual-purpose windings: engine stator and magnetosphere field coil in one set (§7.1). The reverse layout is REJECTED (see §13).

§2.4 The axle service law

- Design the axle AND its service access gaps TOGETHER — not reconciled later. The service route is part of the axle design.

- Axle extensions: allow a few mm of extra diameter for torque twist increasing past the bearing at spin-up. Twist grows with torque and length; THE SPIN-UP TRANSIENT IS THE SIZING CASE, not cruise.

- The turbine does not spin up in place — for the chemical spin, the Long Johns lower each section clear of the hull through its bay doors; after the ~40-minute spin the sections retract inside and the doors close. Full directions §7.1.

§3 — MACHINE TWO: THE LONG JOHNS — THE SUSPENSION BEARING SET (THE POINT HOLDERS)

§3.1 What they are (the Author's dictation, seated)

The end of each turbine is an inverted cone on the end of the centre axle. The Long John arms come down from the hangar bay roof carrying conical points. The cones are both repelling magnets — both north. The rotor floats on repulsion, point to cup, both ends; nothing touches. This is the suspension bearing set that holds the rotor — both in place, and at the exact correct distance and positions from the Quinn coil sets (the turbine's four sets — one per 50 m section). One datum does both jobs: bearing and generator air-gap. Drones move the arms into place; the arms close into the bearing set; they open again for service.

§3.2 Why this shape

The bearing sits at the spin axis — the centrifuge is zero there, and in orbit there is no gravity load on the float gap at all. Contactless: no wear, no lubricant, no particulate in the bay, no stiction when the spin-up pairs light from rest. The cone geometry is a progressive centering spring by the repulsion-stack law (coaxial repulsion rises steeply as the gap closes): any drift off centre closes the gap on that side and the field pushes the point back to the middle — axial spring and radial self-centering in one shaped pair.

§3.3 The stability law — build it honestly

Static permanent-magnet repulsion alone cannot be stable in every mode (Earnshaw's theorem); one axis always runs soft. The closures: the rotor's own spin stiffens the set gyroscopically; the passive cones carry the steady datum (passive is the law); the fine modes close on ACTIVE TRIM COILS — which double as the exact-distance adjusters to the Quinn coil sets — or, to stay fully passive, on superconducting flux pinning (cryo is already aboard for the hydrogen and helium stores). The axle law is honoured: non-ferromagnetic axle; the magnets ride the cone faces, not the axle.

§3.4 The load case — gyroscopic, not gravity

The rotor's angular momentum is $1.106 \times 10^{11} \text{ kg.m}^2/\text{s}$ at the high line (4.17×10^{10} at the 150 MW anchor). Turning the ship at 1 deg/s against it puts a 1.93 GN.m precession moment on the bearing couple — per side now (the geometry law): half the momentum over the 100 m run, the same ~984 tonnes-force per end (98 tf at 0.1 deg/s; 371 tf per end at the cruise anchor at 1 deg/s). Passive capacity: ~14.6 tf per m^2 of cone face at a 0.6 T working gap — a 10 m^2 face holds ~146 tf per end, the steady datum and the gentle turns. SIZE THE TRIM FOR 1,000 tf PER END and the Long Johns hold every legal maneuver. Eddy note: keep the bearing flux out of the generator gap — stray flux taxes the spin as drag; intentional drag belongs to the regen braking through the coils.

§3.5 The name

LONG JOHNS — after Long John Silver, by way of Karl Moeller: sound man, Emmy winner, author of Return to Treasure Island (the book, 2015; ISBN 9780692409046), from whom the Author took the bounce-back idea for the shields; they met online when the Author audited the screenplay Moeller wrote from his book. The mechanism rhymes with the namesake's invention — the U-shaped Perspex holder with a half ping-pong ball on each end: push the sides, the flex opens, the roll pops in. Two shaped halves, and the roll, invented by another man he knew named Yiannis Argyros (John Silver)— the rotor — rides held between points that never touch it.

§4 — MACHINE THREE: THE SPIN-UP PAIRS — CHEMICAL ONLY (THE AUTHOR'S RULING, SEATED LAW)

§4.1 The ruling and its three legs

The rockets on the spin-up engines can only be chemical (the Author's ruling, verbatim): "i've estimated the flux band coming off those engines and they would kill trying to run any sort of plasma nozzle, not to

mention the centrifuge of the spin would displace the contents of the nozzle." Graded TRUE on all three legs. (1) THE FLUX BAND: in vacuum a chemical plume balloons and backflows far beyond its exit cone; the rim line lives inside a dense, hot, partially ionized flux at 4.4-4.5 km/s, and no magnetic nozzle holds its geometry, its drive electronics, or its insulators inside that band — and there is nowhere on the rim outside it. (2) THE CENTRIFUGE: rim acceleration climbs 0 -> 3,983 g across the spin-up (568 g at the 150 MW anchor). At the high line a 30 kg magnet-coil segment would weigh ~119.5 tonnes-force (1,172 kN) and a one-metre LH2 feed column develops 2.77 MPa of centrifugal head before any pump turns — the centrifuge kills structure, feed, cooling, and confinement stability. (3) THE POWER CIRCLE — the decisive leg: the rotor is the battery and it starts EMPTY; an electric nozzle matched to one pair's 10.1 GW jet would want an ~11.9 GW bus that does not exist until the very machine it is trying to spin is already spinning. Chemical breaks the circle: self-powered, self-contained, indifferent to its own exhaust. During a cold spin-up it is the only choice that works at all.

§4.2 The engines — what exists and can be bought

PRIMARY: the RS-25 class (the RS-25E, Aerojet Rocketdyne — the production line restarted, new engines hot-firing through 2025 into 2026): 2,279 kN vacuum at 109%, Isp 452 s, 3.53 t per unit. ONE PER RIM POSITION — pairs on alternate sides at each section end, EIGHT LIVE UNITS in opposed balanced pairs on the full machine (the seated fit): 18.4 MN tangential, full spin in about 10 minutes with all eight live.

ALTERNATES, GRADED: VULCAIN 2.1 (ArianeGroup — 1,370 kN, Isp 431 s, in production for Ariane 6): two per position; full spin 33.6 minutes on one pair-set; 2,617 t. BE-3U (Blue Origin — 890 kN vacuum, in-space restartable by design, flying on New Glenn): three per position; 34.5 minutes; ~2,534 t. RL10 (110 kN, Isp 465.5 s, sixty years of proven in-space restarts): the trim and respin vernier — 42 per pair as the main spinner is cluster madness, but its Isp trims the bill to 2,423 t. REJECTED: Raptor — thrust-perfect (one per position, 34 minutes) but not for sale, and it burns methane, which is not in the store (§13). RETIRED: RS-68A — would have been ideal one-per-position (25.9 minutes) but the line closed with Delta IV in 2024.

§4.3 The spin ledger (seated figures)

Full rotor, rest to the 625 m/s high line: rim impulse 1.106e10 N.s — fixed regardless of how the burn is scheduled. Propellant 2,495 t at Isp 452 (2,506 t at Isp 450): 356 t hydrogen + 2,139 t oxygen at mixture ratio 6:1. Times on the RS-25-class fit: all eight units live — about 10.1 minutes; two pairs — 20.2 minutes; one pair — 40.4 minutes at a 10.1 GW jet (the rim receives ~2.5 GW at the seated 25% rocket-to-rim transfer). PER 50 m SECTION, exactly one quarter: 4,424 t of rim, 240 MWh stored, 624 t of propellant (89 t H₂ + 535 t O₂), 10.1 minutes on one pair — four sections sequentially: the same 40.4 minutes and the same 2,495 t, because the impulse is additive. RESPIN from the 50% speed floor: ~1,253 t. The carried store covers the spin plus one full restart.

§4.4 Scheduling — the impulse is additive and the vacuum coast is free

The burn does not have to be one firing. Options: (a) ONE CONTINUOUS BURN — outside the RS-25's ~8.5-minute flight envelope but a certification extension, not physics: a chamber reaches thermal steady state in the first minute and the design life is measured in hours (heritage: 7.5 h, 55 starts); (b) ONE BURN PER SECTION — 10.1 minutes each, inside the flight envelope, four starts — in-space start for the RS-25 was studied (Ares I) and never flown, so it is a development item; (c) RL10 or BE-3U clusters pulse and restart at will. The fuel is 2,495 t either way.

§4.5 Carriage and mounts

The bill is ship stores, not a stage: 356 t LH2 = 5,025 m³ (a 21.3 m sphere equivalent) and 2,139 t LOX = 1,875 m³ (a 15.3 m sphere equivalent) — orthogrid cryo tankage built at this scale today; it rides inside the ship's stores with the oxygen add. Bottles, never one big tank (the blast physics, seated: one tank goes bang it takes out the ship; the same volume in cylinders blows the path of least resistance across the matrix, and almost always less than half will go). MOUNTS SIZE FOR THE SHUTDOWN STATE: one 3.53 t unit pulls ~2,000 tonnes-force at the 568 g cruise anchor and ~14,000 tf at the 3,983 g high line. Feed runs against the growing centrifugal head — fire from rest upward, as the ledger assumes.

§5 — MACHINE FOUR: THE HYDROGEN FACTORY

- Electrolyser economy (seated): 50 kWh per kg of hydrogen.
- At the 40 MW cruise rating: 800 kg/h = 19.2 t of hydrogen per day.
- At the 150 MW emergency rating: 72 t/day (emergency fill rider).
- FEEDSTOCK — the water ledger, so 'sufficient water' has a number: 9 t of water per tonne of hydrogen.
- BY-PRODUCT: every tonne of hydrogen hands back 8 t of oxygen — oxidizer for the tap engines, air for the crew.

§6 — MACHINE FIVE: THE BABUSHKA STORE

- Cell: nested 'babushka' tanks — bullet-shaped high-pressure cells at 2,300 psi; hydrogen storage density 11.7 kg/m³. Reference cell: the 200 m³ 'house-cell'.
- THE DOLL SPEC (his ruling, seated sitting 856, verbatim core: "4 times a 40foot shipping container for cubic metre space is the largest of the dolls... the others are subject to internal collapsible frames as they get smaller... a radial base of 6m max"): the largest doll carries 4x40-ft-container cubic space — 308 m³ envelope class (~270 m³ usable) — resolved geometry: 5.8 m inside diameter x ~11.5 m on the 6 m max radial base; the smaller dolls nest inside on internal collapsible frames (~0.85 linear step), the scissor-brace doctrine carried inward; a nested ten stow in the big doll's own 308 m³ footprint. Capacities, for fabrication and safety engineering only — under THE JERRY-CAN LAW they never appear as range, reserve, or planning figures: big doll 3.6 t H2 gas at 2,300 psi; nested ten ~9.3 t across 792 m³; liquefied, the whole fleet's catch is ~131 m³ of the eighteenth tank's 16,042 m³. Shell honesty,

bracketed for his materials pick: 2,300 psi on a 6 m radial base is hoop-sized — big doll ~13 t class carbon (1,250 MPa working) / ~123 t class steel (625 MPa); the nested ten ~32.2 t carbon / ~315.5 t steel. Either rides inside the margin. -

THE HANDLING LAW (his ruling, seated sitting 857): the 6 m radial base is the CEILING, never the target — the working gauge is the 2.9 m class, set by handling: a 6 m base-seal disc at 2,300 psi fights ~450 MN of end load (~46,000 tf); at 2.9 m the same closure fights ~105 MN — 4.28x lighter, end load scaling with the square of the base. The fleet may seat as TWO SETS of smaller dolls in place of one deep nest — bench-set by handling class, pod-launch compatibility, the ride up behind the shuttles on the shared logistics gauge, and the available volume at the forward point of the landing/loading bay. The shell bracket does not move: barrel mass per cubic metre is diameter-independent ($m/V = 2\rho$) — the split buys handling and gauge compatibility at zero mass cost; the ends get lighter, the barrel bill is flat. Hard rails: 6 m absolute max, 2.9 m preferred; the jerry-can law governs contents regardless of set count.

- Mechanics (from the seated cell design): internal structural cross-braces scissor FLAT to load a tank, then drop by spring-tension into rigid load yokes; high-density polymer sacrificial buffer sleeves insulate the skins against metal scoring.

- SHIP TANKAGE (the distributed cryo keel, his layout): SEVENTEEN drop-down tanks along the keel, each 15 m long (5 m of it nose cone) x 55 m wide x 25 m drop — spanning ~340 m along the keel. Capacity $272,708 \text{ m}^3 = 19,362 \text{ t}$ of liquid hydrogen at 71 kg/m^3 — the doubled store, grown by three at the seated 1,139 t class (his ruling, 3 September 2026, sitting 827: "add 3" take it up, so we aren't doing this endlessly — the mini tanks made the grow cheap). The compact alternative stays on the shelf: seven units at a 50 m drop, same capacity, ~140 m span — but 50 m out is half the hull height (peggy); the 25 m set drops a quarter-height and feeds the gain: the nozzles run all along the ship. Deployment law stands on every unit: drop on transit, retract for meteor storms or approaching ships. Tank loads are micro-g class — the ship's thrust is $\sim 1\text{e-}5 \text{ g}$, so the tanks hold shape, not weight; structure is his 60-100 t figure (audit bracket 300-800 t at light earth-spec, his ruling governs — the difference does not move the rocket math). The babushka cells ride nested and empty to the gas giants and come home full. SPIN-UP LAW (his ruling): all fuel cryo sections remain UP INSIDE the ship during every spin-up — the keel drops for the cruise legs only, never while the rockets turn the rim.

- COLLECTION ROLE: at the gas giants the babushkas are the bottles — the shearer fleet sweeps, the collection array catches, and the nested cells are filled. Empty cells nest for the run out; full cells stack for the run home.

- THE COLLECTION-STORE LAW (his ruling, seated sitting 854, verbatim core: "we use babushkas to collect space excess, so put in an empty dropdown tank, that will be their unload point... not a cost, a safety price"): the cells live as CRUISE collectors of space excess off the collection array — tap, never fill

(§15.6) — unloading into ONE DEDICATED EMPTY DROP-DOWN TANK, the eighteenth keel tank, their unload point. Price: one more shell at the seated foil-class figure, ~21 t — 1.3% of the ledger margin, no rebalance owed; the cells are already aboard. What it buys: the never-priced dark reserve — every gram the road hands over liquefies into the eighteenth tank, free hydrogen, no electrolysis bill, held separate from the burn store. The keel pays by plan; the eighteenth tank pays by providence. Honest rider: the tap is SLOW (months-to-years class; the rate figure stays owed) — this is a bucket under a drip, not a second keel: whatever accumulates is margin the plan never spent. -

THE JERRY-CAN LAW (his ruling, seated sitting 855, verbatim core: "on earth you go on a long trip, you take a jerry can for emergency, but you never ever price it into the trip... it should not even appear on any ship computer screen"): the eighteenth tank and the whole catch-store are NEVER priced into any trip, burn plan, range figure, or fuel-gauge display — the reserve you count is a reliance, and reliance is the killer; the earthquake box in the basement is not house food. The ship drives on the seventeen, full stop. Lineage: the babushkas were born to exactly this law (the one-puncture value) — the collectors and the unload point were both always jerry cans. Extends the seated reserve law (the reserve is never spent) from the propellant ledger to the energy reserve. Engineering translation, preserving the intent exactly: the tank stays inspectable — safety telemetry (pressure, temperature, relief state) rides the ENGINEERING maintenance channel only; never the fuel displays, never a planning screen, never a range number. The crew who would spend it cannot see it; the crew who must keep it safe can. No MWh from the eighteenth tank may be quoted in any burn plan, payer ledger, or mission profile from sitting 855 forward.

§7 — MACHINE SIX: THE MAIN ENGINE — THE MICROWAVE-HYDROGEN MACHINE

§7.1 The role law (seated)

The main engine throws the Earth departure and the Jupiter run-up on tanked hydrogen — and drives the deep-space legs. It burns PURE HYDROGEN, no oxygen, under microwave fire — the Author's ruling, verbatim: "we can burn hydrogen without oxygen under microwave, hotter but more thrust." One engine design, one propellant: HYDROGEN. The plasma cores are not the climb engine and never were; they are helium machines (§8).

§7.2 The throttle law

At fixed power, thrust $\times$ exhaust velocity = 2P. Per gigawatt of jet power: 167 kN at v_e 12 km/s; 41 kN at v_e 49 km/s. The exhaust-velocity band is I_{sp} 1,000-3,000 (v_e 10-30 km/s); the plasma- v_e seat is 49 km/s.

§7.3 The microwave bank (the big microwave system for the hydrogen — the Author's words)

~470 $\times$ 100 kW 2.45 GHz magnetrons per 40 MW of microwave fire (47 MW electric at the seated 85% chain). GW-class: 500 MW = ~5,882 tubes (~176 t, 88.2 MW waste heat); 1 GW = ~11,765 tubes (~353 t, 176.5 MW waste heat). Waste heat goes to the seated paths. Cooling is not a constraint — we have

space: the fin rig sets, yet to be adopted in the final assembly, will cover this. The same magnetron art as the helium cores (§12.2) — one tube family, two duties.

CAVITY SIDE (the 30 kW cavity class): the verified microwave-hydrogen class is 30 kW per cavity — the 915 MHz NASA Glenn / Penn State / RSI resonant-cavity system, modelled above 60% conversion. So the 40 MW of microwave fire is absorbed by ~1,333 cavities, each 100 kW tube feeding ~2.8 (~800 at 50 kW if the class climbs). The two per-cavity walls are named: plasma reflection at critical density ($7.45 \times 10^{16} \text{ m}^{-3}$ at 2.45 GHz — overdense cores absorb at the critical surface) and thermal dissipation past 30 kW at the quartz jacket. The doctrine already answers both: no wall contact, and count, not size — the physics is demonstrated per cavity and the plant is the count.

THE CLIMB BILL, UNDER THE DISTRIBUTED KEEL: the ship carries 19,362 t of liquid hydrogen in seventeen drop-down tanks (his ruling, sitting 827 — the keel grown three). At the 49 km/s seat the ruled burn — 15,837 t at the full price (the fixed fraction, 12.06% of the 131,315 t departure), 82% of the load, reserve law intact — throws 6.3 km/s: burnout 14.1 km/s, hyperbolic excess 8.8 km/s, aphelion at Jupiter's 5.2 AU, coast to the crossing 2.73 years. THE PATIENT ROAD IS THE RULED ROAD — the keel's depth is what makes the floor legal (the same burn is 99% of the old tanks — a reserve-law breach). The hot option rides the shelf: burning the full burnable 14,350 t threw 7.22 km/s at the seated baseline (aphelion ~9 AU, a 1.4-year coast) but arrives at 11.6 km/s excess — the capture brake wants ~1.60 km/s, about 2,900 t, against 1,595 t aboard; at the grown keel the shelf option burns 17,426 t for ~7.0 km/s and still cannot stop — the hot road is a flyby road. Lower on the band the same tonnage buys less velocity — the band law stands: the higher the exhaust velocity, the more velocity each tonne buys.

§7.4 The departure bill (Earth to Jupiter)

The bill at the 49 km/s seat, on the 104,761 t departure mass: burn 12,639 t for 6.3 km/s — 16.6 years at four modules, and the module ladder compresses it (eight modules halve it). The 10% reserve law binds every leg: never burn past 90% of the load; the ruled burn holds 79% — 3,306 t still aboard on arrival. THE BATTLE MANOEUVRE PAYS THE CAPTURE: the approach is shaped to meet Ganymede tangentially at its 15 RJ orbit — the tail-chase power slide sheds up to ~1.2 km/s against the ~1.0 the capture needs at that altitude; the ship captures on moon momentum onto the long ellipse, then multiple slides on successive laps pump it down toward the 2 RJ park, small propellant trims between slides — the accelerator tap. Arrival cost drops from ~800 t to trim-class; the reserve (1,594 t) rides as pure margin. The moon is the anchor: the momentum dumps into $1.48 \times 10^{23} \text{ kg}$ that never notices — the flown playbook of every moon-tour orbiter. At Jupiter the tanks refill from the giant (~0.8 year) and the tether-fed run-up climbs to 30,000 mph, collecting as it burns (30,676 t, ~6.5 years at the 25-module tether seat).

§7.6 The module throttle ladder (time to speed by module count)

One module = 10 MW electric = 7.2 MW of jet. At the 49 km/s seat a module burns 0.52 t of hydrogen a day; four modules — the cruise plant — burn 2.08 t a day. The departure burn at four modules: 12,639 t in 16.6 years, reserve held; eight modules 8.3 years; the ladder scales by count. Lower on the band the burn is faster and the reach shorter. The GW-class jump stands apart: 1 GW of jet at the 49 km/s seat burns 72 t a day — the ruled departure burn at GW fire is 176 days.

§7.5 The impulse honesty line

Electrified hydrogen delivers ~2.3 kN.s/kg of total impulse against chemical's ~31 — the microwave machine wins on propellant per unit velocity (per carried kilogram at ve 12: 12,000 N.s against the hydrolox mix's 4,433 — 2.7x for carrying no oxygen) and loses on total impulse per kilogram. Size the tanks by the first truth and the calendars by the second.

§8 — MACHINE SEVEN: THE PLASMA CORES AND THE PLASMA ENGINES — HELIUM MACHINES

§8.1 The role law (seated)

The plasma cores are HELIUM machines — the Author's helium-to-plasma invention. Their duties are the shields and the tapping. The helium store is the emergency plasma reserve — 6,000 t liquid in THE SIX-PACK — six dewars distributed along the keel on the drop-down law, each ~8,000 m³ carrying ~1,000 t, never touched unless emergency. There is no open-space collection: the shearer fleet and the collection array refill at the gas giants, at orbital speeds.

§8.2 THE PLASMA ENGINES (the terminal machine)

- SPEC LAW: the engine is everything after the plasma exists (Phase 11 supplementary act, seated law; directions inlined §12.5). One engine design, one propellant — HELIUM: the cores are helium machines; the climb belongs to the hydrogen engines (§1, §5). The cores are the bridge; this machine is the exhaust end.

- THE MAGNETIC NOZZLE — NO WALL CONTACT: shaped coils form a magnetic throat downstream of each core; the plasma expands against the field, never against metal. A physical wall does two kinds of damage at once — eats the wall and freezes the plume; the field does neither. No throat erosion, no ablative liner, no replacement schedule: the wear part is a current.

- THRUST VECTORING WITHOUT MOVING PARTS: steering is differential current — imbalance the nozzle coils, the field throat leans, the plume follows. No gimbal actuators, no hydraulic joints, no flex couplings in the hot path (the same solid-state religion as the PEMS surfaces and the maglev tracks).

- THE ARRAY, NOT THE ENGINE: many small core-and-nozzle units, cross-fed, individually isolated — one unit fails, its neighbours trim, the ship never notices (Phase 37 distributed doctrine). Throttle is by unit count and by microwave power per unit; the fine trim between units is the steering's fine end.

- THROTTLE LAW: two knobs, both electric — microwave power sets plasma density, propellant feed sets mass flow. Response at the speed of a current change — no turbopump spool, no valve lag. Start and shutdown are field events, not ignition events: nothing to light, nothing to purge.
- THE DETACHMENT LAW: the nozzle field is shaped to diverge and weaken downstream so the plasma outruns its own confinement; the cores run pulsed when fine detachment matters — each pulse sheds clean instead of dragging a continuous attached tongue [the pulsed-detachment figures are owed to the bench — Draw Rule applies].
- POWER AND HEAT: the cores draw from the ship grid (Phase 45 air-core alternators); the Quinn turbine chain pays the climb (§2, §11). Waste heat goes to the seated paths — Coldrock dissipation (Phase 9), cryogenic over-sizing (Phase 138): coil cryo is sized against continuous full-power cruise, not average.
- NOZZLE COILS, SPECIFIED (the seated specs): REBCO-class HTS tape running 20-77 K. Each throat carries a swirl-trim coil pair — the Hall swirl plus secondary corrective coils — as the steering's fine end. The vacuum cooling stack: pulse-tube cryocoolers (zero vibration), multi-layer insulation bouncing ~99% of the radiative load, deep-space radiators facing the 2.7 K void — the JWST benchmark chain (five-layer MLI to ~40 K passive, helium pulse-tube to 6.7 K active). THE REGENERATIVE PRIZE: route the 20 K hydrogen feed through the HTS coil jacket before the cavity — free cooling for the coils, free preheat for the feed. Nozzle class demonstrated in adjacent art: VASIMR VX-200SS sustained 200 kW through a magnetic nozzle (RF-helicon on argon — the nozzle proof stands; the 30 kW cavity correction in §7.3 governs the microwave-hydrogen cells).
- BENCH PATH: the 7-second rig grows a scale — same microwave, same bottle, same coiled hose, hung on a torsion arm. Millinewtons per kilowatt on bottled helium first, then the funnel. Owed to the bench: thrust per kW at core temperature; detachment cleanliness pulsed vs continuous; coil current vs vector angle. Until the bench pays them, the figures stay marked (§9; Draw Rule: CARDBOARD > pencil > law text > mesh > render).

§9 — MACHINE EIGHT: THE TETHER CAPTURE ARRAY — THE HARVEST AT THE GIANTS

§9.1 The physics

At Jupiter: arrival v -infinity ~6 km/s; periapsis 42.5 km/s at 2 RJ against co-rotating plasma at 25.1 km/s — EMF ~930 V/km on the pass, ~251 V/km in orbit. A 20 km tether at 500 A takes 535 N of drag and makes 9.3 MW on the pass. The drag is the throttle: the load sets the braking, the braking sets the orbit. (The giant's own generator runs the same physics at scale: the Io flux tube carries ~400 kV, ~5 MA, ~2 TW continuous, fuelled by Jupiter's rotation.)

§9.2 The pools and the calendar

Bare capture: 209 GWh. The full spiral to circular: 5,359 GWh — 15.3 years of the 40 MW book. Time to take it with drag: one tether, 31.6 years; a 100-tether array, ~115 days; a 300-array, ~38 days. The 100-array holds 250 MW orbit-steady and 930 MW on the pass.

§9.3 The chain — capture to spend

Capture: the tether circuit and its power conditioning unit — the load is the throttle. Storage: the flywheel bank (24 cylinders, 2.7-14.4 MWh each), the sodium bank (100-160 Wh/kg; 1,000 t = 100-160 MWh), and the Quinn rotor (960 MWh — one day of cruise). Spend: the magnetron bank (\$7.3) drives the main engine. Surplus AS FUEL: 250 MW spare electrolyses ~120 t of hydrogen a day into the cryo hull (390 kWh/m³).

§9.4 The doctrine lines

Tethers for watts. Station kit never ship kit — the charging-station infrastructure (moon anchors, the big banks) stays at the giant, unmanned only; the ship carries wires, not stations. Mars has no dynamo (~3.5 mV/m) — there is nothing to tap there; the giants are the harvest.

§10 — THE OPERATIONAL VARIANT — THE JUPITER RUN (THE MISSION LAW, SEATED)

§10.1 The sequence

Leave Earth orbit on the hydrogen mains — tanks full, reserve held, the patient road -> coast 2.73 years -> THE BATTLE MANOEUVRE: the shaped approach, the Ganymede power slide at 15 RJ (capture on moon momentum, propellant trims only), then slide-pumping lap by lap down to the 2 RJ park — the first crew break rides the capture ellipse while the slides work -> fill the hulls; THE SHEARER FLEET COLLECTS HYDROGEN FIRST (Jupiter's cloud tops run ~89.8% hydrogen; directions §12.6) -> the crew-sanity stair: return, drop shearers, deploy lasers, drop tethers; run up to 15,000 mph; six-month loop; back, run to 22,000 mph; six-month loop; back, run to 30,000 mph — collecting as we burn in orbit at no energy cost from the tank — and go. The helium cores keep the shields and the tapping the whole way. Departure home rides the helium account.

The departure from Earth orbit under the 79,870 t ship (fully Equipped and bootstrap weight; the rotor rim carried; the distributed keel aboard): 19,362 t of hydrogen in seventeen drop-down tanks, burn 15,837 t at the 49 km/s seat (the ruled fixed fraction at the full price) — 6.3 km/s, burnout 14.1 km/s, hyperbolic excess 8.8 km/s, aphelion at 5.2 AU. THE PATIENT ROAD, RULED: no tether assist owed for the transfer, and the arrival speed — 5.7 km/s excess — is the one the capture doctrine is built on. The burn runs 20.8 years at four modules — the machine does not alter: added stores weight adds time, not fuel or components, and the grown store is what keeps the floor legal under the reserve law.

Departure total, liquid-helium road, distributed keel: 104,761 t at the seated baseline (dry 79,870 t (fully Equipped and bootstrap weight)) — THE CONSOLIDATION (his rulings): the engines move INSIDE the

ship; the four turbine sections ride in internal bays, two port and two starboard; the stator coils seat in the outer hull wall as dual-purpose windings (engine stator and magnetosphere field coil in one set); the magnetosphere rides in those hull windings — the hull is 500 x 180 x 100 m; an internal magnetic shield wall stands between the engine bays and the crew spaces; bay doors below each section let the Long Johns lower the engines clear of the hull for the rocket spin-up and retract them after the ~40-minute spin; no tugs, no positioning fleet; all fuel cryo sections remain up inside the ship during every spin-up. THE PRICE, STATED HONESTLY: the widening adds +96,000 m² of skin (220,000 → 316,000 m²) = +19,636 t on the seated skin law; the forward frame section adds +1,000 t of steel — frame and forward allocation, +3,400 t total; the stator copper doubles to 80 t across two full sets. THE KEEL GROWS — HIS RULING, SEATED (3 September 2026, sitting 827: "add 3" take it up, so we aren't doing this endlessly): +3 drop-down tanks at the seated 1,139 t class — SEVENTEEN tanks, 19,362 t of hydrogen aboard, ~340 m of keel span; dry ≈ 103,007 t (the three added shells ride at his foil-class figure, ~21 t — inside the carried bracket, immaterial to the rocket math); departure reads ≈131,315 t; the floor burn 15,837 t (the ruled fixed fraction — 12.06% of departure) against 17,426 t burnable — margin +1,589 t and the 6.30 floor holds exactly — LEGAL AT THE FULL PRICE, the fork CLOSED. THE TEETH otherwise stand: 12,639 t buys exactly 6.30 km/s at the seated baseline (a steering target, not a worry: sixteen years of throttle to hit it, and the slide margin is fattest there). Thrust at the cruise plant, ve 49: four modules = 40 MW electric = ~29 MW of jet ≈ 1.2 kN. Conversion honesty: 40 MW electric makes ~29 MW of jet (85% x 85%) — the calendars are already x1.4 the ideal.

§10.3 The capture burn

THE BATTLE MANOEUVRE PAYS THE CAPTURE (the Author's design, ruled): the approach is shaped so the periapsis sits out at Ganymede's 15 RJ orbit and the ship meets the moon tangentially — the tail-chase power slide sheds up to ~1.2 km/s of Jupiter-centred speed against the ~1.0 the capture needs at that altitude, and the ship captures onto the long ellipse on moon momentum alone. Then the speedway laps: slide after slide, pumping the ellipse down toward the 2 RJ park, small propellant trims between slides — the accelerator tap. The moon is the friction surface. The Sunday drives avoid banking because banking pays momentum; the battle manoeuvre banks on purpose — the same exchange, aimed. The reserve — 1,594 t, 0.87 km/s at the 2 RJ periapsis — rides as pure margin behind the slide. The first crew-sanity loop is the capture ellipse itself.

THE SLIDE MATHS, IN FULL — BENCH-GRADE, AS DIRECTED: the moon is a 1.48×10^{23} kg anchor and the flyby is momentum exchange with it — flown physics: the Galileo orbiter's tour, Europa Clipper, JUICE. Oberth cuts both ways: the cheap 0.43 km/s capture brake is the deep-periapsis price at 2 RJ; the slide happens high at 15 RJ where the ship moves 16.4 km/s, so the slide must shed ~0.99 km/s. The envelope: nose-in geometry gives 0.37 km/s — short, the ship leaves hyperbolic; the shaped tail-chase approach gives up to 1.19 km/s — capture closes, thin, and thin is enough. The hot road is dead to the slide: it needs 3.49 km/s at the slide point against a best-case 0.83 — flyby-only, confirmed twice. THE

WINGS RULING IS MISSION-CRITICAL: the light disc stack (7,850 t the pair) keeps the departure burn at 6.68 km/s — the floor cleared with margin; the heavy end (15,700 t) drags it to 6.20 km/s — below floor, the ship misses the crossing. The Jupiter stop runs ~8.8 years all-in: fill ~0.8, run-up ~6.5, loops 1.5. The hot road stays shelved as flyby-only: the brake wants ~2,900 t against 1,595 t aboard — the ship cannot stop on it. The bench cannot hold a moon — this one is proved by the flown record, not the rig.

§10.4 The helium account

The helium store is 6,000 t in THE SIX-PACK — six dewars distributed along the keel on the same drop-down law as the hydrogen tanks, each ~8,000 m³ (~25 m sphere class) carrying ~1,000 t — with a cryocooler per dewar, plant ~440 t all-in. The emergency plasma reserve, never touched unless emergency. Leg draws run 720-840 t across the campaign; the store comes home at ~5,200 t. The plasma cores are helium machines; the six-pack is their only aboard feed. There is no Jupiter helium collection.

§10.5 The additions law

This variant adds exactly TWO systems to the base design (the Author's words): the tether capture systems, and the big microwave system for the hydrogen. Nothing else changes.

§11 — THE DEPARTURE CALENDAR (EMPTY TO DEPARTURE, WITH SUFFICIENT WATER)

FILL TIMES at the 40 MW cruise rating (19.2 t/day):

Cryo hulls, 19,362 t 1,008 days (THIRTY-THREE AND A HALF MONTHS)

Spin-fuel hydrogen share, 356 t 19 days

At Jupiter the hull refills from the giant, not the factory (~0.5 year) — station kit never ship kit

Departure sequence: spin up and refill before leaving — the tank leaves full, the spin bottles charged, solar helping the margin

EMERGENCY RIDER at the 150 MW rating (72 t/day):

Cryo hulls, full load 221 days

WATER LEDGER: the ship does not carry water for hydrogen — the hydrogen flies as hydrogen, in the cryo hull, filled at Earth and refilled at the giants. Water aboard is life support only — people, oxygen,

and food production — still a high store, carried inside the 20,000 t outfit allowance and sized by crew and cycle, not by propellant.

OXYGEN RETURN: none owed to propellant — the oxidizer-return line belonged to the carry-water doctrine. The crew's oxygen comes from the life-support loop.

THE STRANDED SHIP'S ANSWER, COMPLETE: water in, departure out,
ten and a half months — with the machine that spins the whole
chain already paid for.

§11B — THE SPIRAL TRANSIT (THE DUTY PROFILE)

- THE PATH: the ship leaves the solar system on a continuous outward spiral and carries that same spiral out into the galaxy (the Author's order). An always-under-drive profile — the low-thrust electric character of the cores is the trait the path is built on: the ship climbs on patience, not fire (§12.5).

- THE ARRIVALS STAND: Phase 133's elliptical flower loops govern every destination arrival, unchanged — wide low-energy loops around the destination body, velocity kept, infrastructure dropped, no dead-stop parking orbit, no braking-burn waste. The spiral governs the transit; the flower governs the arrival.

- THE DUTY: the helium store is a 50-year store at the seated average duty of 20 GWh/yr — 2.3 MW average against the 40 MW cruise feed. The ship does not burn full drive continuously on a spiral; it climbs, cruises, and loiters. No top-ups are needed in the first 50 years.

- THE SECOND 50: two-month stays at helium-rich environments are the balancing metric (the Author's order). The shearer fleet does the sweeping (§12.6); the collection array does the catching (§12.4); the bottles take the bank (Phase 11 §11-§12).

§12 — CROSS-PHASE DIRECTIONS, INLINED (PROGRAM LAW: DIRECTIONS, NOT POINTERS)

§12.1 THE INTERNAL ENGINE BAYS (the turbine lives inside the ship)

Current law for the engine installation:

- THE ENGINES ARE INSIDE THE SHIP (the Author's ruling): the four 50 m turbine sections ride in internal bays along the hull — two port, two starboard, one clockwise and one counter-clockwise per side, the not-twisted-ship law held pair-wise — 10 MW per section at the ladder's own unit size, 40 MW all four.

- THE COILS ARE THE OUTER WALL (the Author's ruling): the stator coils are seated in the outer hull wall as DUAL-PURPOSE windings — engine stator and magnetosphere field coil in one set — the

magnetosphere rides in those hull windings. An INTERNAL MAGNETIC SHIELD WALL stands between the engine bays and the crew spaces against the engine fields.

- SPIN-UP DEPLOYMENT (the Author's ruling): bay doors below each engine section; the Long Johns lower the sections clear of the hull for the rocket spin-up and retract them inside after the ~40-minute spin. No tugs, no positioning fleet — nothing leaves the ship but the sections themselves, on their own bearings. All fuel cryo sections stay up inside during every spin-up.

- THE STORM LAW, INTERNAL: four separated, shield-walled bays — a hit on one side cannot take the engine; any two sections still make 20 MW, and the full four hold 40 MW at the seated speed.

- THE FORWARD FRAME SECTION: +1,000 t of steel, chassis-linked, at the forward lower deck between the cryo rooms.

- THE GEOMETRY LAW (the Author's ruling — no wing maths anywhere): the turbine run is 100 m PER SIDE — two 50 m sections port, two starboard; the 200 m is the whole machine across both sides, never a per-side run. THE RETRACTION LAW (the Author's ruling): after spin-up the spun sections retract UP into the outer-hull coil set, and the inner hull set swings down/across and locks in place — the stator closes around the rotor, outer set and inner set, and the ship runs with the engines home.

- THE COPPER WEIGHT (the generator maths, the Author's delegation — 'not my call, thats the known maths for a generator'): at the seated numbers the rim speed does the work — 40 MW across 12,566 m² of rotor surface at 625 m/s wants a shear of only 5.1 Pa; at a 1 T gap field and 5 A/mm² the physics floor is ~115 kg of copper for the whole set (~860 kg at 150 MW emergency, 0.5 T). The carried figure is the industrial line — 10 t per section, 40 t the set: end turns, leads, cooling manifolds and fault duty — the ledger carries the copper DOUBLED, 80 t across two full sets, immaterial to the rocket math, seated so the ledger closes.

- THE COMPLETENESS NOTATION (the Author's ruling): THE STARSHIP GALILEO IS COMPLETE — every space contingency covered, the NASA roadblocks gone. Nothing about this ship waits on invention.

§12.2 From PHASE 11 — THE AMBIENT HELIUM MICROWAVE PLASMA PROPULSION CORE (the helium cruise cores — never the climb; the climb rides the main engine, §7)

- CORE PHYSICS: 2.45 GHz microwave excitation of helium to plasma — electrodeless breakout.

Verification baseline (the Author's bench test, seated verbatim in the record): one microwave, a bottle of party-gas helium, a drill, a coiled hose (about 8–10 inches, 6 rings) — gas on, microwave on, plasma in UNDER 7 SECONDS.

- CONSTRUCTION PATTERN: helium runs continuously through internal COILED REACTION CHAMBERS bombarded by the high-output 2.45 GHz field; the plasma is expelled through ELECTROMAGNETIC NOZZLES. No electrodes in the plasma path (the durability point); the field is the pipe.
- OPERATING POINT (seated): Isp 5,000 — exhaust ~50 km/s. At the 13.4 km/s (30,000 mph) cruise against a 50 km/s exhaust, catch-and-throw is NET THRUST (the ram tax: net momentum per kilo = exhaust speed minus ship speed; the sign flips only past 50 km/s ship speed).
- COLLECTION LAW: there is no open-space collection and no speed threshold. Collection happens at the gas giants, at orbital speeds — the shearer fleet sweeps the cloud tops and the EM array catches (§12.6). Between stops the ship lives on its stores; at the stops the stores refill. PROPELLANT SPLIT (the Author's ruling): the HYDROGEN ENGINES throw the ship on TANKED HYDROGEN from the cryo hull while the plasma cores run the shields and the tapping on the HELIUM store. The cores are never fed hydrogen.
- RAM DOCTRINE, the four catches (so the collector is built honestly): (1) magnets only grab charge — a forward ionizer (laser or electron beam) must strip neutral gas ahead of the throat; (2) the ram tax, above; (3) mirror loss — a converging field reflects a share of ions back out; throat geometry decides the fraction; (4) density is the tyrant — raw interstellar medium gives grams per year; the collector is the NET, the seeded lane is the STOCKED RIVER. The highway buoys stock the river; Phase 242's mini-moons are lane lights, never power or propellant components.
- ORIENTATION NOTE (the renames): early Phase 11 text cites power 'from the Phase 3 Kinetic Turbine' — that machine is THIS phase's Quinn turbine. Current Phase 3 is the bootstrap stores (§12.3). An alternate exhaust routing feeds the Phase 21 plasma stealth cloak.

§12.3 From PHASE 3 — THE BOOTSTRAP LAUNCH STORES (what the chain runs on before the collector lights)

- Carry initial onboard HELIUM for the plasma and HYDROGEN in the cryo hull (plus Nitropril). The bootstrap stores bridge the ship from launch to the first gas-giant stop; the chain (this phase) is what spends and refills them.

§12.4 From PHASE 3, AMENDMENT 1 — THE EM COLLECTION ARRAY (the HOW of collection at the gas giants)

At the gas giants the cores top up on harvested helium and the stores refill (§1, §12.2). THIS is the machine that does the harvesting — seated law from Phase 3 directions inlined in full per the program law. The aboard stores (§12.3) carry the ship between stops; the array is what refills them at the giants.

- THE TRICK — the funnel is a field, not a structure. A physical scoop can only be as wide as what you build; an EM array cheats — superconductor loops project a magnetic field kilometres wide, and charged particles that cross it spiral down the field lines into a throat metres wide. A 10 m loop wears a 2 km invisible net. Effective collection area = the field's cross-section, not the ship's. Proven physics

running free tonight: Earth's magnetosphere funnels solar wind into the poles — an aurora is a planet running a collection array.

- THE COILS (Phase 228 hardware, consolidated): the field coils ARE THE OUTER WALL — the stator set is built into the hull's outer wall and dual-purposed as the magnetosphere, driven-mode and segmented, tuned from the closed shield-bubble geometry to the OPEN FORWARD FUNNEL. One hardware family, engine drive and shield field both. (228's original mercury-ring drive stands with the Author's own caveat seated: mercury is inert until a hull breach — non-inductive superconductor is the preferred coil path; weight and breach-safety decide.)

- THE IONIZER (Phase 249 hardware): magnets only grab charge — raw neutral gas does not know your field exists and flies straight through the prettiest funnel ever wound. The laser rides FORWARD of the ship (side-center external mount; band receptor discipline for the oscillation wander) and strips the gas far enough ahead that the fresh ions have room to curve into the throat. It is a power bill — that bill is why most paper ramjets die — and this phase's turbine is what pays it.

- THE RAM TAX: everything you catch arrives at ship speed and must be re-thrown; net momentum per kilo = exhaust speed minus ship speed. At 30,000 mph (13.4 km/s) against the 50 km/s helium exhaust the fleet is on the winning side — catch-and-throw is net thrust: the cores pay the energy, the medium supplies the mass, the tanks refill. Cross 50 km/s ship speed and the same array becomes a brake. Cruise collection speed's honest definition: BELOW EXHAUST SPEED.

- THE MIRROR BITE: a converging field reflects a share of incoming ions straight back out (pitch-angle mirroring); you never collect everything that enters the net — throat geometry decides the fraction. Design knob, not a killer.

- THE DENSITY TYRANT: run the wall math — 30,000 mph through a 2 km funnel in raw interstellar medium (about 1 atom/cm³) collects a few grams per year. The array does not beat the wall; it never did. The array is the NET — the seeded lane (highway buoys) is the STOCKED RIVER. Net without river is jewellery. Build for the lane. At the gas-giant stops the shearer fleet does the sweeping and this array does the catching (§12.6).

- NOT A NEW MACHINE: the array is a field geometry on seated parts — 228's coils re-tuned open, 249's laser re-tasked forward, the mule plasma engines as the thrower. The ship grows its scoop out of hardware it already flies.

§12.5 From PHASE 11 — SUPPLEMENTARY ACT: THE HELIUM PLASMA ENGINES (the terminal machine this chain drives)

The chain's cores hand their plasma to the engines: shaped coils form a MAGNETIC THROAT downstream of each core — the plasma expands against the field and never touches metal. No throat erosion, no ablative liner, no replacement schedule: the wear part is a current. Steering is differential

current — imbalance the nozzle coils and the field throat leans, the plume follows: no gimbal actuators, no hydraulic joints, no moving parts in the hot path.

The ship flies an ARRAY, not an engine: many small core-and-nozzle units, cross-fed, individually isolated — one unit fails, its neighbors trim, the ship never notices (same doctrine as Phase 37's distributed rail thrusters). Throttle is two electric knobs: microwave power sets plasma density, propellant feed sets mass flow; response at the speed of a current change, and start and shutdown are field events — nothing to light, nothing to purge. The nozzle field diverges and weakens downstream so the plume detaches clean; the cores run pulsed when fine detachment matters [the pulsed-detachment figures are owed to the bench — Draw Rule applies].

One plasma, three jobs: the engines thrust; the Phase 21 cloak bleeds the same plasma through the Phase 13 AION hull ports; the Phase 76 branch is the same physics pointed. Bench path for the owed numbers: the 7-second rig hung on a torsion arm — thrust per kW at core temperature, detachment cleanliness pulsed vs continuous, coil current vs vector angle. Until the bench pays them, the figures stay marked.

§12.6 THE SHEARER FLEET — THE STOP-TIME COLLECTION MACHINES

- ROLE: forward drones flying LOWER than the ship at the gas-giant stops, shearing the dense fringe upward into the ship's lower entrance, where the EM collection array (§12.4) catches. Grader-and-bulldozer doctrine: the shearer pushes the line over; the ship catches it. Running lower is physics, not preference — the push needs ion-neutral coupling, and the coupling lives deep: mean free path 10 km at $10^8/\text{cm}^3$ (leaks), 1 km at 10^9 (works), 100 m at 10^{10} (grabs hard).

- THE BLADE: a one-sided 45-degree thin-film deflection sail. A 1 km vertical peel is 1.414 km² of 12-micron polyimide-class film — about 24 t per blade plus rigging. A dozen blades cover a 16.97 km² spread for ~290 t of film fleet-wide.

- WHY SAIL, NOT BEAM: at stop densities the gas cannot be dodged and does not need ionizing — at $10^{10}/\text{cm}^3$ its self-collision mean free path is 0.4 km: it behaves as a wind, it has mass, and it does not go through the sail. Impact energy at the 16 km/s graze is 1.3 eV per hydrogen atom (5.4 eV helium) — below the 20-30 eV sputter threshold, so the film cannot be sandblasted; heating runs 0.34-34 W/m² across 10^8 - 10^{10} , radiated away. The sail pushes neutrals directly — 100% of what hits it — and pays no ionization tax.

- THE THRUST BILL (seated honestly): shear duty is continuous-thrust duty — Newton's third law is the mechanism; the push the gas gets is the push the blade feels. Per blade per 60-day stay: at $10^9/\text{cm}^3$, 0.43 kN reaction -> 23-45 t propellant and 10-21 MW of drive power; at 10^{10} , 4.3 kN -> 226-453 t and 105-210 MW. Deploy and return legs are trivial.

- THE SELF-FEED CLOSURE: the blade stands in its own fuel stream. Tap ~16% of the sheared flow at the blade root, run it through plasma cores at I_{sp} 10,000, and the shearer eats its own peel — the propellant bill goes to net zero. Root ingestion at 16 km/s relative is owed bench physics.
- THE BUFFER: each drone carries a ~1 GWh battery buffer (2,000-4,000 t Li-class), recharged from the ship by umbilical or beam between passes. Cells age like bottle seals — the buffer is replaceable hardware and rides the spares doctrine (Phase 11 §12).
- ENVELOPE: sweet spot 10^9 - 10^{11} /cm³. At 10^{12} the thrust bill climbs to 428 kN per blade — the ceiling region. Below 10^8 the sweep is thin — that work belongs to the beam mode (ionizer plus funnel at standoff). Two-mode doctrine: SAIL in the dense, BEAM in the thin.
- DELIVERY NUMBERS: a dozen blades at 10^{10} /cm³ sweep 11.6 t/h of gas toward the entrance (helium ~1.4 t/h) — about 2,000 t of helium crossing the blades per 60-day stay. At a 10% catch fraction that banks ~200 t of helium per stay. The entrance catch cap is the remaining gate (Phase 11 §11).
- BENCH OWED: root ingestion at 16 km/s; film hydrogenation class, seaming, rigging dynamics, tear-out; the concentration factor C — peel area into entrance area — measured, not spoken.
- LASER ROLE UNDER THIS DOCTRINE: the arrays are planetary scoops — they run at the stops, on catch duty at the lower entrance, never as a continuous cruise burn (a constant wide curtain is terawatt-class dead physics).

§13 — REJECTED OPTIONS AND DEAD ENDS (DO NOT BUILD)

- REJECTED — magnets on the wings / stator on the cylinder: 'makes giant unmanageable stators we would struggle with.' Rotor carries magnets; stators stay small and racked.
- DEAD END — costing the chemical climb on water: the 149,400 t figure costed the whole water molecule — 16,600 t of it is hydrogen at the 9:1 split. The climb is done by THE HYDROGEN ENGINES — the main engine — on tanked hydrogen; the plasma cores are HELIUM machines and are never fed hydrogen.
- REJECTED — the 200 MW emergency rating: 150 MW is FINAL (the no-redlines ruling). Do not design to 200.
- CONSTRAINT — never size a field that eddy-heats the carbon rim: field size reflects carbon impedance (§2.3).
- REJECTED — physical or ablative nozzles on a core-temperature plume: the wall loses twice — it erodes, and it freezes the plume. The nozzle is a field, not a pipe (§8.2). DO NOT BUILD.
- REJECTED — the single big bell: one engine is one failure point and one concentrated torque point — the outboard-motor stupidity (the ship spreads the load; §12 doctrine). The ship flies an array. DO NOT BUILD.

- REJECTED — gimballing thrusters: a hinge in the hot path is a leak, a lag, and a maintenance debt. Steering is differential current. DO NOT BUILD.
- REJECTED — plasma nozzles on the rim for spin-up: the flux band kills the confinement, the centrifuge kills the structure and the feed, and the power circle makes it impossible — the rotor is the battery and it starts empty (§4.1). DO NOT BUILD.
- REJECTED — electric or microwave spin-up variants: priced (922 t at ve 12 km/s, 553 t at ve 20) and rejected on the Author's ruling. The spin-up pairs are chemical (§4). DO NOT BUILD.
- REJECTED — methalox on the rim (Raptor class): thrust-perfect but not for sale, and methane is not in the store — a second propellant system for one job. DO NOT BUILD.
- RETIRED — the RS-68A option: one per position, 25.9 minutes — but the production line closed with Delta IV in 2024. Unbuyable; do not specify.

§14 — VERIFICATION CHECKS (CHECK YOUR WORK AGAINST THESE)

TURBINE

full spin $625 \text{ m/s} = 596.8 \text{ rpm} = 62.5 \text{ rad/s}$ on the 20 m-diameter rim | period 0.1005 s | rim accel 3,983 g | 960 MWh stored

carbon hoop @ $625 \text{ m/s} = 625 \text{ MPa}$ vs 1,250 MPa ceiling -> margin 2.0x (ship-side)

steel hoop @ $236 \text{ m/s} = 436 \text{ MPa}$ vs 1,000 MPa ceiling -> margin 2.3x (colony-side steel anchor, the 150 MW point)

carbon rim ceiling = 884 m/s (7.9x at emergency spin)

FACTORY

50 kWh/kg -> 40 MW gives $800 \text{ kg/h} = 19.2 \text{ t/day}$; 150 MW gives 72 t/day

water : hydrogen = 9 : 1 ; oxygen return = 8 t per t H₂

STORE

2,300 psi nested cells; hydrogen 11.7 kg/m³ (babushka collection bottles)

17 drop-down cryo tanks (15 m + 5 m nose x 55 m wide x 25 m drop) = 272,708 m³ = 19,362 t LH₂ |

~340 m keel span | nozzles feed all along the ship

ENGINES

RS-25 class: 2,279 kN vac, Isp 452 -> 36.8 mm/s² on 61,850 t

CALENDAR

830 d hull fill | 19 d spin-fuel share | 16.6 yr Earth burn at four modules

221 d emergency fill | ~8.8 yr Jupiter stop with the crew-sanity stair

PLASMA (§12.2)

2.45 GHz, sub-7-second breakout, electrodeless, EM nozzles

Isp 5,000 / ~50 km/s exhaust | collection at the gas giants, never open space

SHIELDS (§12.1)

4 x 50 m internal turbine sections (2 port + 2 starboard, CW + CCW) | dual-purpose stator coils in the outer wall | 10 MW per section | 40 MW total | forward chassis frame section +1,000 t steel (forward lower deck, between the cryo rooms)

209 m/s rim | ~344 MPa vs ~690 MPa plate | two float-bearing sets

PLASMA ENGINES

torsion-arm thrust stand: millinewtons per kW on bottled helium, logged vs microwave power

detachment: plume sheds clean, pulsed vs continuous — no attached tongue dragging thrust home

vector: coil-current differential vs plume angle, mapped and logged; nothing moves in the hot path

array: kill one unit at cruise — neighbours trim, attitude holds

SHEARER FLEET (§12.6)

blade: 1 km peel at 45 deg = 1.414 km² film = ~24 t + rigging | a dozen = 16.97 km² spread, ~290 t film

thrust bill per blade per 60-day stay: 0.43 kN / 23-45 t prop / 10-21 MW @ 10⁹ | 4.3 kN / 226-453 t / 105-210 MW @ 10¹⁰

film survival: 1.3 eV H / 5.4 eV He at 16 km/s < 20-30 eV sputter threshold | heat 0.34-34 W/m²

self-feed: ~16% of sheared flow at Isp 10,000 -> net-zero propellant (bench-owed)

HELIUM STORE (Phase 11 §11-§12)

6,000 t He liquid in THE SIX-PACK — six keel dewars, each ~8,000 m³ (~25 m sphere class) carrying ~1,000 t, ~48,000 m³ at 125 kg/m³, 4.2 K | lowers on transit, retracts for storms and approaching ships the 50-year duty at 0.33 t/day average | zero-boiloff socket ~50-100 kW | emergency plasma reserve, never touched unless emergency

THE THROTTLE LAW: $F \times v_e = 2P \rightarrow$ at 1 GW and v_e 12 km/s: $2 \times 1e9 / 12,000 = 167$ kN. Check §7.2.

THE SPIN-UP IMPULSE: $J = m_{rim} \times v_{rim} = 17,695,000 \times 625 = 1.106e10$ N.s, radius-independent. Fuel = $J / (452 \times 9.807) = 2,495$ t = 356 t H₂ + 2,139 t O₂ (MR 6:1). Check §4.3.

THE PER-SECTION QUARTER: 624 t, 10.1 min on one pair, 240 MWh — the impulse is additive; four sections sequential = the full-rotor totals. Check §4.3.

SPIN TIME IS RADIUS-INDEPENDENT: $t = m_{rim} \times v_{rim} / F_{tangential}$. One pair (4,558 kN): 40.4 min; two pairs: 20.2; the seated eight-unit fit (18.4 MN): about 10.1. Check §4.2-4.3.

THE TETHER EMF: ~930 V/km on the 2 RJ pass (42.5 vs 25.1 km/s), ~251 V/km in orbit; 20 km at 500 A = 535 N drag, 9.3 MW. Check §9.1.

THE LONG JOHN LOAD: $L = I.w = 1.7695e9 \times 62.5 = 1.106e11$ kg.m²/s (the whole machine; 5.53e10 per side); a 1 deg/s turn (0.01745 rad/s) $\rightarrow$ 1.93 GN.m machine-wide, 9.65e8 N.m per side; over the 100 m per-side couple = 9.65 MN = 984 tf per end — the momentum and the arm halve together, the per-end load is unchanged; trim sized 1,000 tf. Check §3.4.

THE DEPARTURE CLOSES THE STORE: the ruled burn — 12,639 t, 79% of the load — at the 49 km/s seat throws 6.3 km/s, burnout 14.1, excess 8.8 — the patient road, aphelion at Jupiter's 5.2 AU. Check §7.4, §10.2.

§15 — THE POWER ACCOUNTING

§15.1 THE ANCHORS — THE HIGH LINE, OURS (the ugly-storm rule: we run the high numbers as ours). Rim speed 625 m/s = 596.8 rpm = 62.5 rad/s on the 20 m-diameter rim ($r = 10$ m); rotation period 0.1005 s; rim acceleration 3,983 g. Effective rim mass 17,695 t across the 20 m diameter (uniform-shell equivalent ~0.88 m). Stored energy 960 MWh — 54.25 kWh/t. Hoop stress 625 MPa against the 1,250 MPa working ceiling — margin 2.0x, no redline. Service windows: full discharge 24 h at 40 MW, 6.4 h at 150 MW; to the 50% speed floor (720 MWh usable) 18 h at 40 MW, 4.8 h at 150 MW. Lenz reaction torque: 0.64 MN.m at 40 MW, 2.40 MN.m at 150 MW — drawing power slows the rotor, and the slowdown IS the fuel gauge (Phase 249 doctrine).

§15.2 CITED AS HOPEFUL UNDER BEST DESIGN (his ruling): the Grok low line — 55-90 t wheel at 236 m/s, ~540 kWh stored, 3,000 kg per spin, 6,000 kg aboard (Isp 280 s, transfer efficiency 0.4). His note seated verbatim: Grok's estimates say the system will run better than ours. If the mass build lands

light, the low line becomes the lived reality and the margin comes home as spare store; if it lands heavy, the ship already carries the storm.

§15.3 THE SPIN ENGINE (his machine): the rotors are cut into four 50 m sections; rocket-spun, alternate sides on each end, respun as required — and NOT an endless task: once at speed they are not spun again unless the ship slows. At the landing bay the plenum actuators extend outside and pin-lock each end during spin-up — balance only, because rockets never fire exactly the same at the start; torque is a pure couple through the centre of the drive, little to none on the ship. **THE OUTER WALL CARRIES THE COILS** — the stator is the hull's outer wall, dual-purpose with the magnetosphere field. Centering the spin inwards; spin collars for the actuators (+4-8 t allowance). **REGENERATIVE BRAKING** through the same induction path — controlled stop in minutes-to-hours, then pin-lock, then the crew swaps bottles; no weeks of coast-down. — spin-up with the sections lowered clear of the hull through the bay doors, cryo sections stowed inside

§15.4 THE SPIN FUEL — MAX-BUDGET STORE (the worst cost is the carried cost): rim impulse to full spin 11.06 GN.s, fixed regardless of burn time. High line: 2,495 t per full spin at Isp 452 (2,506 t at Isp 450; RS-25-class) — 356 t hydrogen + 2,139 t oxygen; per 50 m section 624 t; respin from the 50% speed floor ~1,253 t; the spin-up pairs are **CHEMICAL ONLY** (\$4); the carried store covers the spin plus one full restart. Eight live RS-25-class units in opposed balanced pairs gives 18.4 MN and about 10 minutes to full spin; a 60-second spin would want ~80 such units live. Bottles, never one big tank: many small removable cylinders in a spaced, vented matrix — his blast physics seated: 1 tank goes bang it takes out the ship; the same volume in cylinders blows the path of least resistance across the matrix, the bottles blow each other away, and almost always less than half will go. The tank shells add weight; the labour saving and the survival case carry it. **UPRATE LADDER (FOUR-UNIT SPLIT-LAW RIDER, his ruling):** spin propellant scales linearly with rim speed above the seated 625 m/s high line — $1.1x = 2,745$ t; $1.25x = 3,119$ t; hard ceiling $1.41x = 3,528$ t, where the 884 m/s tip meets the carbon hoop's 1,250 MPa working ceiling exactly. The working band is 1.1-1.25x (+250-624 t over the seated store); the exact figure seats with his speed number. **THE TEETH:** speed alone cannot replace a lost section — 2x the 625 line wants a 1,250 m/s tip, past the carbon ceiling; the storm law holds by section count (§12.1), not by overspeed.

§15.4a THE REFILL LAW (seated sitting 847, his challenge: "how were you proposing to fill rockets travelling at 300ms even at half speed"): **NOBODY FILLS A ROCKET AT SPEED** — the doctrine never proposed it; his own build-doc law ruled it at the seating: "we need to stop them to safely refuel rockets." The burn fires **FROM REST UPWARD:** feed runs through the spoke/axle against the growing centrifugal head (2.77 MPa per metre at the high line — pump-sized, flagged and seated), and at ignition the rim is at zero; the 300-plus m/s only exists because the burn already happened. Once at speed the sections are not spun again unless the ship slows — not some endless task. When a respin is owed, the slowdown IS the refill window by design: **REGENERATIVE BRAKING** through the induction

path stops a section in minutes-to-hours (his order — no weeks of coast-down), then pin-lock at the bay, then the crew swaps the bottle matrix — full removable cylinders in from ship stores, empties out, the spaced vented matrix carrying his blast physics. The empties refill at cruise off the collection arrays — tap, never fill (§15.6). Four sections, serviced one at a time; the carried store covers the spin plus one full restart. THE SEATED SEQUENCE, SIX WORDS: slow, brake, pin, swap, spin, depart.

§15.4b THE DUMP LAW (seated sitting 925, his rulings verbatim, 7 September 2026: "Ok the 50% of the bottle store obviously gets used at ship launch to speed, the second 50% is held until we have run 6 months no main tank issues, and dump it in the spare tank or used tanks for the primary engines, it is flammable i prefer it outside of the ship. Assign bottle dump and refill conveyers to a line point connection, meaning we can control dump to a chosen tank, and refill at that same point if needed, bottles were originally because the engines were outside. A secondary high value compressor will be needed, as the tanks will be higher pressure than the bottle, refill is easy dumping will need mechanics.") — THE 50/50 LAW: the carried bottle store is two halves by doctrine — the first half (one full spin, 2,495 t) spins the ship at launch; the second half (the restart reserve) holds to the six-month no-main-tank-issues gate, then dumps into the spare tank or the emptied primary tanks. THE OUTSIDE LAW: the residual spin fuel does not linger aboard — it is flammable; it moves into the tank farm. THE LINE POINT LAW: the bottle dump and refill conveyors ride a line point connection — dump steerable to a chosen tank, refill at the same point if needed; one point, two directions, chosen tank. THE COMPRESSOR DEBT: a secondary high-capability compressor joins the bill — the receiving tanks sit at higher pressure than the bottles, so refill runs downhill and the dump runs uphill into pressure and needs the mechanics; gas-phase dump is the seated reading, a liquid-phase pump prices separately at bench. PROVENANCE, HIS: the bottle matrix was born when the engines rode outside; the engines are inside now (§15 STATUS), the bottles stay — their role is this law. THE STATION BOILOFF FLAG CLOSES BY THIS RULING: the bottles never hold cryo for years — half burns at spin-up, half dumps at the gate into the managed tanks.

§15.4c THE RED LINE LAW (seated sitting 926, his rulings verbatim, 7 September 2026: "The spec change leaves the same allocated and math fuel on board, so the secondary respin is still there, we will have to calibrate tank fuel gauges to red line that as main drive empty so it doesnt get used as burn fuel, or counted as burn fuel") — THE ALLOCATION LAW: the dump moves the fuel, not the math — the same allocated tonnage stays on board under the same numbers (2,495 t, one full respin from rest), and the secondary respin capability stands intact after the dump. THE RED LINE: the tank fuel gauges are calibrated to read the respin allocation as MAIN DRIVE EMPTY — the red line sits above the reserve, so the drive can never draw it and no planning display or burn ledger ever counts it as burn fuel. THE FAMILY OF THE PROTECTED: the third walled-off quantity aboard — beside the 10% reserve law (never burned) and the jerry-can catch-store (invisible to every planning display and fuel

ledger) — the respin allocation is the rotor's fuel riding in the drive's tanks, marked and walled off by the gauge law.

§15.5 THE PAYER, NAMED: the Quinn Turbine is a flywheel, not a source. The payers in the dark are the cryo tanks — 19,362 t H₂ is ~645,000 MWh thermal, ~322,000 MWh as fuel-cell electric: about eleven months of the 40 MW cruise feed — and as propellant through the jet, twenty-five years of cruise burn. In-system, the perpendicular slide-out solar arrays on the hull flanks pay build-phase and cruise loads. Gas-giant stops refill hydrogen without the electrolysis bill. NEVER draw power without a seated payer — Lenz reaction torque is continuous; uncompensated generation is a brake (Phase 44 law). — the array mounts on the hull (perpendicular slide-out panels)

§15.5a THE DECAY TABLE — THE WHEEL AS PAYER, PRICED (seated sitting 849, his words: "the energy from the machine runs the microwaves that turn the hydrogen to plasma, thus the draw must have some slowing effect" — TRUE, and it is the seated Lenz law: every amp drawn is counter-torque; the slowdown IS the fuel gauge): if the wheel ever carried the 40 MW feed alone, E falling with the square of rim speed: one hour — 625.0 to 611.8 m/s (-13.2); six hours — 541.3; twelve — 441.9; EIGHTEEN HOURS — the 50% floor at 312.5 m/s, where a respin costs ~1,253 t; twenty-four — empty. That is the wall the doctrine never leans on: rockets as cruise payer would be one full spin per day (~2,506 t/day) — forbidden. The wheel is the buffer and the gauge, never the payer: the seated payers (arrays in-system, cryo store as fuel-cell electric in the dark, the Babushka store, the gas-giant refills) feed the magnetrons direct and refill the wheel continuously, so the rim speed HOLDS while the books balance second by second; when they do not balance, the wheel pays at this table, the falling rim is the alarm, and the storm law plus one carried restart stand behind it.

§15.6 TAP, NOT FILL: the collection array is the tap, never the fill — collection rates are slow, and the run to full speed may take months to years. The ship leaves with full stores, hydrogen AND helium; the stops top up. The density tyrant stands: the 2 km net in raw interstellar medium collects grams per year. The seeded lane (highway buoys — the 242 mini-moons are lane lights, never power or propellant components) is the stocked river. The ram tax stands: catch-and-throw is net thrust, never net energy.

§15.7 THE MASS LEDGER (seated so every later chapter balances): dry ship 79,870 t (fully Equipped and bootstrap weight) at the seated baseline — the consolidation adds +19,636 t of hull skin — the 61,850 t build (including the 20,000 t outfit allowance: landers, factories, tethers, the plasma plant, machines, cannon, crew water and all bootstrap stores) PLUS the rotor rim, 17,695 t, ruled additional and now carried, PLUS the distributed tank set at his 100 t figure (audit bracket 300-800 t, immaterial to the rocket math) and the helium six-pack plant delta ~225 t; PLUS the forward frame section — +1,000 t of steel, chassis-linked, at the forward lower deck between the cryo rooms; PLUS the stator copper, 80 t across TWO full sets (the generator maths, seated at the industrial line — the physics

floor is ~115 kg, the rim speed does the work); THE CONSOLIDATION (his rulings): the engines move INSIDE the ship — four 50 m sections in internal bays (two port, two starboard, CW+CCW per side, 10 MW per section), stator coils dual-purpose in the outer wall (engine + magnetosphere in one winding), the magnetosphere riding in the outer-hull windings — hull 500 x 180 x 100 m (+96,000 m² of skin = +19,636 t on the seated skin law) — with an internal magnetic shield wall between engine bays and crew spaces, bay doors below each section, the Long Johns lowering the engines clear for spin-up and retracting them after the ~40-minute spin; no tugs; all cryo sections stowed inside during spin-ups. The wings as separate structures are dissolved — no housings, no dock, no deployment, no armour. NEW MASS STATE — FULL PRICE, THE KEEL GROWN (his ruling, sitting 827 — "add 3" take it up, so we aren't doing this endlessly): dry ≈ 103,007 t (the three added tank shells ride at his foil-class figure, ~21 t, inside the carried bracket); departure ≈ 131,315 t; the floor burn reads 15,837 t (the ruled fixed fraction — 12.06% of departure) against 17,426 t burnable (the reserve — 1,936 t — is never spent) — margin +1,589 t and the 6.30 km/s floor holds exactly — LEGAL AS SHE STANDS, the fork CLOSED. The mini tanks made the grow cheap: same 25 m drop class, the keel span runs ~280 m to ~340 m. At the seated baseline the ruled burn stays 12,639 t = exactly 6.30 km/s. Hydrogen 19,362 t in seventeen drop-down keel tanks (his ruling, sitting 827 — the fourteen-tank keel grown by three at the seated 1,139 t class); helium 6,000 t in six dewars plus ~440 t plant; spin-fuel store 2,495 t per \$15.4, uprated under the split law per the \$15.4 ladder (the exact figure seats with his speed number) plus bottle-shell mass; Sea Witch 11.2 t with ammunition. Departure total: 104,761 t at the seated baseline; ≈131,315 t at the full price — the keel grown three tanks (sitting 827). The rim question is CLOSED — the rim was never in the dry figure and is now carried. — THE EIGHTEENTH TANK (his ruling, sitting 854): one dedicated EMPTY drop-down rides as the babushka unload point for cruise collection — shell at the foil-class figure ~21 t, inside the +1,589 t margin, no rebalance owed; it fills as the road provides and is never counted in the burn ledger.

\$15.7a THE ROCKET-CLASS MASS RIDER (seated sitting 847, his question: "did we/anybody account for the weight of the class of rocket we chose to rim weight output?"): graded honestly — the rim (17,695 t) and the propellant (2,495 t plus one full restart) were carried; the hardware was not, until now. (i) THE UNITS: eight live RS-25-class at 3.53 t = +28.2 t, seated herewith. (ii) THE MOUNTS: flagged at the seating (sized for the SHUTDOWN state — one unit pulls ~14,000 tonnes-force at the 3,983 g high line) but never priced — bracket 5-15 t per unit, +40-120 t. (iii) SPARE SETS: the build-doc doctrine carries multiple spare rocket sets — two spare sets +56.5 t. (iv) THE BOTTLE SHELLS: named above ("plus bottle-shell mass") but never numbered — bracket ~375-625 t at crane-class 10 m³ cylinders to ~1,000-1,300 t at crew-carry 1 m³ class; it hangs on the standing owed flag THE SPIN-BOTTLE VESSEL SPEC, now promoted: the bottle size class is a ledger ruling, his to speak. THE RIM-OUTPUT HALF: mass at the rim is not dead weight against output — every tonne at 625 m/s stores 54.25 kWh, so the hardware pays its own way in battery; the only rim costs are the linear impulse growth (tenths of a

percent) and the mount load path (already flagged). MARGIN CHECK: the +1,589 t margin absorbs the low end whole and survives the high end thin (~+85 t) — the ship stays LEGAL on every branch.

§15.8 THE BONUS OPTION: the Mayernick rim wall is not in this system and never was — a bonus option, seated in the master only (his ruling).

§15.9 THE STOP COLLECTORS: the collection at the gas-giant stops is flown by the SHEARER FLEET — a dozen thin-film sail drones flying lower than the ship, shearing the dense fringe up into the lower entrance where the collection array catches (full directions §12.6). The bay-ship dive ship — a smaller ship, five times the lander — remains the heavy option for deep fringes; its design phase is owed, not assumed. The primary reduces speed, de-spins its angular momentum, and orbits at a safe distance while the fleet works.

§15.10 THE NOZZLE LAW (his rulings, 3 September 2026, sitting 828) — the 100-jet array and its plumbing, closed. (1) THE SURFACE LAW: the jets ride as close to the circumference as possible — they are angled, and an angled jet mounted inboard pushes off-centre; at the hull surface the thrust line stays honest. (2) THE 45 LAW: the jets mount in alternating 45-degree pairs pushing rearward to push her forward — the sideways push cancels pair by pair, the axial components add. The audit's price, printed: the cant costs $\cos 45 = 0.707$ of each jet's thrust along the axis — 29% spent for the surface mount and the balanced ring. (3) THE EXCLUSIONS (his words stand): none in the nose crush point; none on the very rear of the ship — a crippled ship strikes the tail, and that is not a panel-beating repair, it is a stranding event; drop-down jets are legal but rear-pointing ONLY, and NONE under the hull — the keel lives there, for obvious reasons. The keep-clear zones seat the failure-mode doctrine: the jets ride only where the ship stays repairable. (4) THE RING LAYOUT: four rings around the hull excluding the base, over the centre 400 m of the 500 m hull — 150 m of ring run, the 100 m engine gap, 150 m of ring run — 300 m of ring-bearing length divided by four = 75 m apart centres, equally set around the top 3/4 arc. On the seated skin-law perimeter ($2 \times (180+100) = 560$ m) the top three-quarter arc is 420 m: 25 stations per ring at 16.8 m centres — about six pod bays at the 3 m seam pitch. Ring stations on the hull: 87.5 / 162.5 / 337.5 / 412.5 m from the nose — 75 m apart within each run, 175 m across the gap, 50 m clear at nose and tail. THE SPLIT, SEATED BY THE LAYOUT: 4 rings x 25 = the 100-jet total = 25 jets per rotor section — his rings seat the audit's flagged natural split exactly. THE ODD STATION, DISCLOSED: 25 is odd — the audit seats 12 alternating 45-degree pairs plus the top-dead-centre station riding straight-rear (the natural spine slot, and where a rear-only jet belongs); his to amend. (5) THE PLUMBING LAW: fuel lines run top and bottom wherever possible; ALL tanks feed cascade to the main lines; TWO full sets, seated NOT CLOSE — and one set rides EMPTY as the meteor-strike spare: a dry line cannot leak, and one strike cannot kill both sets. The empty set is not a saving — it is a second ship's-worth of pipe carried dry. (6) THE POD LAW: the jets live in side pods and follow the same line-ejection sequence already seated for the hull-mounted plasma nozzles of the stealth shield (the seam-bleed law, Phase 21) — ejection self-sealing collar locks, compressor-fitting

class: when the repair matrix throws a damaged pod the line post disengages and goes out with the pod, self-sealing behind it; the same fitting disengages for maintenance. No open line is ever left in the ship. (7) THE COUNT THAT FOLLOWS: 100 HTS nozzle jackets, four rings of 25 — the 20 K feed routing now prices per jet, per ring. (8) THE BALANCE LAW (his ruling, 3 September 2026, sitting 830): the four-ring layout is the SEATED WORKING MODEL — "i just needed a working model no issues and its done" — and more bands may come ("will at some point"). The trade is named both ways in his words: too many in one area is egg-in-a-basket, with the high-stress points a concentrated array cuts into the skin; and too many bands is too many lines of hull-skin openings across the ship. "So i will configure a balance" — the configuration is his, when he wants it. The audit's flag for that day, disclosed: stagger the arc positions band to band so no straight longitudinal line of openings runs the skin — the manifolds prefer alignment, the skin prefers stagger; his balance will seat it.

STATUS: CHAPTER GP-IM-001, v5.25 — 11 SEPTEMBER 2026 (number corrected per sitting 883: the seated chain reached v5.20; the sitting-864 retraction is an amendment, not a rollback — the number never goes backward) — the Author's rulings stand: THE PATIENT ROAD IS THE RULED ROAD (burn 12,639 t = 6.3 km/s at the seated baseline, excess 8.8, coast 2.73 years — legal under the reserve law) and THE BATTLE MANOEUVRE PAYS THE CAPTURE (the shaped approach, the Ganymede power slide at 15 RJ, slide-pumping down to the 2 RJ park — moon momentum, propellant trims only; the reserve rides as pure margin). The hot road (7.22 km/s, 11.6 km/s arrival) is shelved as flyby-only. THE CONSOLIDATION, SEATED: the engines are INSIDE the ship — four 50 m sections in internal bays (two port, two starboard, one CW + one CCW per side; 10 MW per section, the ladder's own unit); the stator coils are the outer hull wall, dual-purpose engine-and-magnetosphere windings; the magnetosphere rides in the outer-hull windings; the hull is 500 x 180 x 100 m; an internal magnetic shield wall guards the crew spaces from the engine fields; bay doors below each section — the Long Johns lower the engines clear for the rocket spin-up and retract them after the ~40-minute spin; no tugs, no positioning fleet; all fuel cryo sections stay up inside during every spin-up. THE RENAME, SEATED (his ruling): the engine is THE QUINN TURBINE — the wing era is over, the machine is named. THE GEOMETRY AND RETRACTION, SEATED (his rulings): 100 m of turbine per side — two 50 m sections port, two starboard, no wing maths anywhere; after spin-up the sections retract UP into the outer- hull coil set and the inner hull set swings down/across and locks in place. THE COPPER WEIGHT, SEATED (the generator maths, his delegation): physics floor ~115 kg the set at the seated numbers — the rim speed does the work; carried at the industrial line, 10 t per section, 40 t the set — and DOUBLED, 80 t across two full sets. THE COMPLETENESS NOTATION, SEATED: the Starship Galileo is

complete — every space contingency covered, the NASA roadblocks gone. THE FORWARD FRAME SECTION, SEATED: +1,000 t of chassis-linked steel at the forward lower deck between the cryo rooms. THE PRICE, IN THE OPEN — FULL PRICE, THE KEEL GROWN (his ruling: "'add 3" take it up, so we arent doing this endlessly', sitting 827): +19,636 t of skin, +3,400 t of frame and forward allocation, +80 t of copper, and +3 drop-down tanks at the seated 1,139 t class — SEVENTEEN tanks, 19,362 t of hydrogen aboard, ~340 m of keel span; dry ≈ 103,007 t; departure reads ≈131,315 t; the floor burn 15,837 t against 17,426 t burnable — margin +1,589 t and the 6.30 floor holds exactly — LEGAL AS SHE STANDS, the fork CLOSED. THE CALENDAR LAW, STATED: at the 40 MW plant (2.08 t H₂/day, ~1.2 kN) the calendar is tonnage-driven — the ruled burn (12,639 t at the seated baseline) runs 16.6 years at four modules; the full burnable load (14,350 t — the reserve is never spent) runs 18.9 years, no matter the ship's mass; at the full price with the grown keel (his ruling, sitting 827) the ruled burn (15,837 t) runs 20.8 years at four modules and the full burnable load (17,426 t — the reserve is never spent) runs 22.9 years; the mass only sets what the tonnage buys (top speed at Earth departure: 33,600 mph at the seated baseline). Departure 104,761 t at the seated baseline, ≈131,315 t at the full price — the keel grown three tanks (sitting 827) — with the frame section and the doubled stator copper aboard. Propulsion is done; THE NOZZLE LAW, SEATED (his rulings, sitting 828, §15.10): the 100-jet array rides the hull surface in four rings of 25 around the top three- quarter arc — alternating 45-degree pairs rearward, the 150-100-150 layout over the centre 400 m, none in the nose crush point, none at the tail, none under the hull; cascade feed to twin separated mains, one set carried EMPTY as the meteor spare; side pods on the stealth- shield line-ejection sequence; the four-ring layout is the seated working model (sitting 830) — more bands his to balance later: concentration is egg-in-a-basket and stress points, proliferation is too many lines of skin openings. The §15.4 spin law is CLOSED (sitting 924): the high line, the uprate ladder, and the hoop ceiling all seated — the rung pick is a contingency order, not a debt. Current doctrine only, per the clean law: no old designs, no superseded terms, no dated patch-riders — a manual is what a builder builds from. — v5.12 (sitting 847): §15.4a THE REFILL LAW and §15.7a THE ROCKET-CLASS MASS RIDER seated; the spin-bottle vessel spec promoted to a ledger ruling (his to speak). — v5.13 (sitting 849): §15.5a THE DECAY TABLE seated. — v5.14 (sitting 854): THE COLLECTION-STORE LAW seated — the babushka cells live as cruise collectors unloading to the eighteenth drop-down; the dark reserve priced as a safety price. — v5.15 (sitting 855): THE JERRY-CAN LAW — the catch- store is invisible to every planning display and fuel ledger; safety telemetry rides the engineering channel only. — v5.16 (sitting 856): THE DOLL SPEC seated — big doll 308 m3

on a 6 m radial base, collapsible-frame nesting; the cell-volume owed flag CLOSED. — v5.17 (sitting 857): THE HANDLING LAW — 6 m ceiling, 2.9 m working gauge, two-set option, bench-set. — v5.22 (sitting 925): §15.4b THE DUMP LAW seated — the 50/50 bottle law, the outside law, the line point connection, the compressor debt; the station boiloff flag CLOSED by ruling; the stale spin-speed owed line withdrawn (closed sitting 924). — v5.24 (sitting 926): §15.4c THE RED LINE LAW seated — the dump moves the fuel, not the math; the respin allocation red-lined as main-drive-empty on every gauge, never burned, never counted.

THE GEOMETRY RESOLUTION, STATED ONCE FOR THE RECORD: earlier text carried a 200 m-diameter ring reading (59.68 rpm, 398 g rim). The seated design is the 200 m long x 20 m diameter cylinder, $r = 10$ m — the Author's rulings: "50 metres sections", "20m not 2m", and the Long John axle ends. So: $625 \text{ m/s} = 596.8 \text{ rpm} = 62.5 \text{ rad/s} = 3,983 \text{ g at the rim}$; the 150 MW anchor = $225 \text{ rpm} = 236 \text{ m/s}$. Impulse, fuel, stored energy, torque, and spin time are radius-independent and stand unchanged.

PROVENANCE: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it changes by seating, not by editing.

Short Version: 1 GW — JUMP class, (not the ship). 11,765-tube bank, 138 modules', that does the climb in 26–36 days instead of years — the output of the engine rotors, exhaust velocity higher than our maximum. It exists in the record as a priced option for GW-class installations (colony-side, or a ship built to jump), not as the Galileo's fit.

So the ladder reads: 30 kW per cavity (physics) → 100 kW per tube (hardware) → 10 MW per module (packaging) → 40 MW cruise plant (the ship, 4 modules) → 1 GW jump class (the fast option, priced but not fitted).

One sentence version: tubes make 100, cavities eat 30, modules bundle 10 MW, the ship carries four, and the gigawatt is what you'd build if you wanted the climb in days instead of years.

END OF CHAPTER 01.